

ASSESSMENT 2 REPORT: AUTOMATED POTHOLE DETECTION AND SEVERITY GRADING USING TRANSFER LEARNING

**MCTA 4363
DEEP LEARNING**

SECTION 1

SEMESTER 2 25/26

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METHODOLOGY

3.1 Overview

This section presents the methodology developed to achieve the objectives of this study, which include comparing pothole detection and severity classification model across multiple methods, to establish a proposed dataset for pothole severity classification and evaluating system effectiveness for improved road safety. By using three different methods, we compared their performances to determine which method is the best to use in detecting the pothole.

3.2 Dataset & Future Works

The initial dataset that we've used to prepare the models was a publicly available pothole detection model sourced from GitHub compiled by Noor Khokhar. The dataset contained 7233 images total, with 6962 images for training and 271 for validation. The images were said to be collected from various sources across the internet, which cover diverse road conditions and environments. The dataset was made to be used for an object detection model only, as the label provided only detects potholes and does not classify them.

The final goal for our project is to improve upon his dataset with ground truth for a severity classification model for the potholes. The goal is to annotate a quarter of his model to try to get a good enough accuracy for the model to classify the pothole semi-consistently.

3.3 Proposed Methods

This study proposes three distinct methods for pothole detection and severity classification. Each method employs a different technique to analyse pothole

characteristics, enabling a comparative evaluation of their effectiveness.

Method 3.3.1 utilises a bounding box-based approach combined with GrabCut segmentation to isolate pothole regions from the input images. The extracted regions are then processed to obtain relevant features, which are used for pothole classification.

Method 3.3.2 utilises instance segmentation based on YOLOv8-seg inference to identify potholes and extract key features for severity assessment. By analysing detection confidence and size-related attributes, the method enables classification of potholes into three hazard categories.

Method 3.3.3 employs a two-stage heuristic CNN for pothole analysis. The method incorporates distance estimation and depth analysis techniques to extract key physical characteristics of the pothole. These features are then used to classify the pothole into three levels of severity.

3.3.1 Method 1: Grabcut Segmentation

The first method employs bounding box-based detection and combines with Grabcut segmentation to isolate the pothole. In the first stages, potholes are identified using the bounding box method. The bounding box defines the region of interest, ROI, that potentially contains potholes.

Following detection, the GrabCut algorithm is applied within the bounding box to refine the segmentation of the pothole. GrabCut is an iterative segmentation technique that separates the foreground from the background by modelling pixel distributions. Once the pothole region is segmented, the extract region would estimate its severity by using the true size score and the depth-related

score. These scores are then combined to assess the severity of the pothole and classify it into three predefined categories, which are minor, moderate and severe.

3.3.2 Method 2: Two-Stage Heuristic CNN

This method divides road damage analysis into two specialised stages: locating the potholes and applying mathematical rules for grading their severity. In the first stage, the Convolutional Neural Network acts strictly as a scout. It scans the incoming video, identifies the presence of a pothole, and isolates the exact area by drawing a bounding box around it. If no pothole is detected, the system simply loops back and moves to the next frame to save processing power.

Once the bounding box is established, the second stage takes over using hardcoded physics formulas known as heuristics (trueSize and shadowMath). It first calculates the true physical size of the pothole based on its distance from the camera, and then estimates its depth by measuring the dark shadows trapped inside the cropped image. The system finally combines these size and depth metrics into a Hazard Score, automatically classifying the damage as a Minor, Moderate, or Severe threat before displaying the result on the screen.

3.3.3 Method 3: Instance Segmentation

The last method makes use of the basic instance segmentation model from YOLOv8 to capture the pothole shape. The model was trained with the same dataset as per the other method, with 100 epochs, which produced a high-accuracy segmentation model. It will output a mask that isolates the exact pothole region to get more accurate size and depth estimates.

Also, similar to other methods, the severity will be obtained from the size score and the depth score. True size score uses the exact pixel mask of the pothole and will take into consideration the distance of the pothole from the camera. The depth score, meanwhile, uses standard deviation to take a sample of the brightness of the pixel in the pothole. This will then determine the depth by looking at the contrast. The higher the standard deviation, means a deeper the hole due to the difference in the shadows.

3.4 Severity Classification Framework

The framework categorises the severity of detected potholes into three actionable tiers based on a unified Hazard Score, evaluated on a standardised scale from 0 to 100. A score between 0 and 40 designates a Minor Hazard, representing shallow potholes or minor surface anomalies that pose a negligible risk to vehicular stability. A score ranging from 41 to 70 triggers a Moderate Hazard classification, representing more pronounced pothole formations that may cause noticeable passenger discomfort, trigger premature suspension wear, or require minor evasive action from the driver. Finally, a score between 71 and 100 classifies the anomaly as a Severe Hazard, characterising a massive pothole or deep structural crater that presents a critical, immediate danger to vehicle integrity, often necessitating urgent driver intervention to prevent damage. This standardised numerical matrix ensures that, regardless of the underlying artificial intelligence architecture, all models in this study are evaluated against the same strict definitions of pothole danger.

3.5 Evaluation Metrics

To rigorously compare the effectiveness of the three distinct detection methods, this research employs standard

object detection evaluation metrics. Because the models were trained on a specialised, custom dataset annotated with ground truth severity classifications (Minor, Moderate, Severe), the evaluation measures how accurately each model learned to localise and classify these specific physical danger tiers. The comparative analysis relies on four primary statistical metrics: Precision, Recall, F1-Score, and Mean Average Precision (mAP).

Precision is utilised to measure the accuracy of the models' positive predictions, quantifying the percentage of detected anomalies that were truly road hazards. Recall assesses the models' sensitivity, calculating the proportion of actual ground truth potholes successfully identified by the system. To evaluate the balance between these two metrics, the F1-Score is calculated, providing a single metric that represents the harmonic

mean of Precision and Recall. Finally, Mean Average Precision (mAP) serves as the most comprehensive benchmark for spatial and categorical accuracy. The mAP evaluates the precision across all recall values and severity classes while accounting for the Intersection over Union (IoU) of the bounding boxes, ultimately determining which of the three models provides the most reliable spatial localisation and severity prediction.

3.6 Sources

noorkhokhar99/data-traning. (2025). GitHub.

<https://github.com/noorkhokhar99/data-traning>